SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I **K.YAMUNA-192511013** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

K.YAMUNA

Signature of the student:

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

1.A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

File Transfer Using Transport and Data Link Layers

Given

- File Size = 150 MB = 150,000,000 bytes
- Bandwidth = 50 Mbps
- Segment Size = 1500 bytes
- Data Link Layer Overhead = 40 bytes/frame

Formula

Number of Segments

$$\text{Segments} = \frac{\text{File Size}}{\text{Segment Size}}$$

Total Overhead

$$\text{Overhead} = \text{Frames} \times \text{Overhead per Frame}$$

Transmission Time

$$\text{Time} = \frac{\text{Total Bits}}{\text{Bandwidth}}$$

Calculation

Number of Segments

$$\frac{150{,}000{,}000}{1500} = 100{,}000$$

Since one segment becomes one frame,

$$\text{Frames} = 100{,}000$$

Data Link Layer Overhead

$$100{,}000 \times 40$$
$$= 4{,}000{,}000 \text{ bytes}$$

Total Data Sent

$$150{,}000{,}000 + 4{,}000{,}000$$
$$= 154{,}000{,}000 \text{ bytes}$$

Convert into bits

$$154{,}000{,}000 \times 8$$
$$= 1{,}232{,}000{,}000 \text{ bits}$$

Transmission Time

$$\frac{1{,}232{,}000{,}000}{50{,}000{,}000}$$
$$= 24.64 \text{ seconds}$$

Result

- Segments = **100,000**
- Frames = **100,000**
- Data Link Overhead = **4 MB**
- Transmission Time = **24.64 seconds**

Explanation

The Transport Layer divides the file into manageable segments and ensures reliable delivery through sequencing, acknowledgements, and retransmissions. The Data Link Layer encapsulates each segment into frames and performs error detection using CRC and frame checking mechanisms.

Application

In real-world file transfer systems such as cloud storage and enterprise networks, both layers work together to ensure data reaches the destination accurately and efficiently.

Conclusion

Reliable communication is achieved through segmentation, framing, error detection, and retransmission mechanisms, resulting in successful transfer of large files

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2.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Sliding Window Protocol

Given

- Window Size = 6 frames
- Total Frames = 48
- One frame lost per window

Formula

$$\text{Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

$$\text{Retransmissions} = \text{Windows} \times 1$$

Calculation

$$\frac{48}{6} = 8$$

Windows = 8

$$8 \times 1 = 8$$

Retransmissions = 8

Result

- Transmission Windows = 8
- Retransmissions = 8 frames

Explanation

Flow control regulates the amount of data sent before receiving acknowledgements. The sliding window mechanism allows multiple frames to be transmitted simultaneously, improving throughput.

Application

TCP uses sliding window flow control to achieve efficient and reliable communication across the Internet.

Conclusion

Flow control enhances efficiency by reducing idle time and preventing receiver overload.

3.A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

IP Fragmentation

Given

- IP Packet = 12,000 bytes
- MTU = 1500 bytes
- Header = 20 bytes

Formula

Payload per Fragment

$$MTU - Header$$

Number of Fragments

$$\frac{PacketSize}{Payload}$$

Calculation

Payload

$$1500 - 20 = 1480$$

Fragments

$$\frac{12000}{1480} = 8.108$$
$$= 9 \text{ fragments}$$

Header Overhead

$$9 \times 20 = 180$$

Result

- Fragments = **9**
- Header Overhead = **180 bytes**

Explanation

The Network Layer performs fragmentation whenever packet size exceeds MTU limitations. Each fragment receives its own header and is reassembled at the destination.

Application

Fragmentation enables communication across heterogeneous networks with different MTU sizes.

Conclusion

The Network Layer guarantees successful packet delivery by fragmenting, routing, and reassembling packets.

4.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Sliding Window Protocol and Flow Control

Given

- Sliding Window Size = 6 frames
- User Data per Frame = 1024 bytes
- Total Frames Transmitted = 48 frames
- One frame is lost in every window and must be retransmitted

Formula

Number of Transmission Windows

$$\text{Number of Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

Total Retransmissions

$$\text{Retransmissions} = \text{Number of Windows} \times \text{Lost Frames per Window}$$

Calculation

Step 1: Calculate Number of Transmission Windows

$$\text{Number of Windows} = \frac{48}{6}$$
$$= 8$$

Therefore,

$$\boxed{\text{Number of Windows} = 8}$$

Step 2: Calculate Total Retransmissions

One frame is lost in each window.

$$\text{Retransmissions} = 8 \times 1$$
$$= 8$$

Therefore,

$$\boxed{\text{Total Retransmissions} = 8 \text{ frames}}$$

Result

Parameter	Value
Window Size	6 frames
Total Frames Sent	48 frames
Number of Transmission Windows	8
Lost Frames per Window	1
Total Retransmissions	8 frames

Explanation

The Sliding Window Protocol is a flow control mechanism used by the Transport Layer to improve communication efficiency. Instead of waiting for

an acknowledgment after sending each frame, the sender can transmit multiple frames within the window size before receiving acknowledgments.
In this problem, the sender transmits 48 frames using a window size of 6 frames. Therefore, the transmission is divided into 8 windows. Since one frame is lost in every window, one retransmission is required per window, resulting in a total of 8 retransmissions.

Application

Sliding window flow control is widely used in protocols such as TCP (Transmission Control Protocol). It helps networks achieve higher throughput by allowing continuous transmission of data while ensuring that the receiver is not overwhelmed. The protocol also improves reliability by detecting lost frames and retransmitting them when necessary.

How Flow Control Improves Communication Efficiency

1. Prevents Receiver Overload: Ensures that the sender does not transmit data faster than the receiver can process it.
2. Improves Throughput: Multiple frames can be sent before waiting for acknowledgments, reducing idle time.
3. Reduces Network Congestion: Controls the amount of data entering the network.
4. Supports Reliable Communication: Lost frames are detected and retransmitted, ensuring accurate delivery.
5. Efficient Utilization of Bandwidth: Keeps the communication channel active and productive.

Conclusion

Using a sliding window size of 6 frames, the sender requires 8 transmission windows to send 48 frames. Since one frame is lost in each window, a total of 8 retransmissions are required. Flow control improves network performance by preventing congestion, increasing throughput, and ensuring reliable communication between sender and receiver.

5.A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of check points created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronisation in the session layer.

Session Layer Synchronization

Given

- Video File = 600 MB
- Checkpoint Interval = 60 MB
- Failure at = 390 MB

Calculation

Checkpoints:

60, 120, 180, 240, 300, 360

Number of Checkpoints:

$$\frac{360}{60} = 6$$

Data to Retransmit:

$$390 - 360 = 30$$

Result

- Checkpoints Created = 6
- Retransmission Required = 30 MB

Explanation

Synchronization points enable communication recovery from the last successful checkpoint rather than restarting the entire transfer.

Application

Used in multimedia streaming, database transactions, and large file transfers.

Conclusion

Synchronization significantly reduces recovery time and network resource consumption.

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6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Presentation Layer Compression and Encryption

Given

- Original Size = 900 MB
- Compression = 40%
- Encryption Overhead = 5%

Calculation

Compressed Size

$$900 - (900 \times 0.40)$$
$$= 540MB$$

Encrypted Size

$$540 + (540 \times 0.05)$$
$$= 567MB$$

Percentage Reduction

$$\frac{900 - 567}{900} \times 100$$
$$= 37\%$$

Result

- Compressed Size = 540 MB

- Encrypted Size = 567 MB
- Reduction = 37%

Explanation

The Presentation Layer compresses data to reduce bandwidth requirements and encrypts information to ensure confidentiality.

Application

Used in HTTPS, SSL/TLS, multimedia systems, and cloud storage.

Conclusion

Compression and encryption together improve both efficiency and security.

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7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

FTP Application Transfer

Given

- Data = 3 GB = 3000 MB
- Request Size = 3 MB
- Throughput = 120 Mbps

Calculation

Requests

$$\frac{3000}{3} = 1000$$

Data in Megabits

$$3000 \times 8 = 24000Mb$$

Time

$$\frac{24000}{120} = 200s$$

Result

- Requests = **1000**
- Transmission Time = **200 seconds**

Explanation

FTP operates at the Application Layer and provides file transfer services, authentication, and file management.

Conclusion

Application Layer protocols ensure reliable and user-friendly access to network services.

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8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

End-to-End Delay

Given

Layer	Delay
Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms

Layer	Delay
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Calculation

Processing Delay

$$4 + 3 + 5 + 2 + 1 = 15ms$$

Total Delay

$$15 + 18 + 12 = 45ms$$

Result

Total End-to-End Delay = 45 ms

Explanation

Each OSI layer contributes processing time before the packet is transmitted across the network.

Conclusion

Combined processing, transmission, and propagation delays determine overall communication performance.

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9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Network Efficiency

Given

- Useful Data = 80 MB

- Frame Size = 1600 bytes
- Overhead = 80 bytes

Calculation

Payload

$$1600 - 80 = 1520$$

Frames

$$\frac{80,000,000}{1520}$$
$$= 52,632$$

Overhead

$$52,632 \times 80$$
$$= 4,210,560 bytes$$

Efficiency

$$\frac{80,000,000}{84,210,560} \times 100$$
$$= 95\%$$

Result

- Frames = **52,632**
- Overhead = **4,210,560 bytes**
- Efficiency = **95%**

Explanation

Protocol overhead consumes network bandwidth and reduces effective throughput.

Conclusion

Minimizing unnecessary overhead improves network efficiency.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Healthcare IoT Network

Given

- Medical Record = 240 MB
- Compression = 30%
- Segment Size = 1400 bytes
- Frame Overhead = 60 bytes

Calculation

Compressed Size

$$240 - (240 \times 0.30)$$
$$= 168MB$$

Compressed Bytes

$$168{,}000{,}000$$

Segments

$$\frac{168{,}000{,}000}{1400}$$
$$= 120{,}000$$

Overhead

$$120{,}000 \times 60$$

$$= 7{,}200{,}000 bytes$$

Total Data

$$\begin{aligned} &168{,}000{,}000 + 7{,}200{,}000 \\ &= 175{,}200{,}000 bytes \\ &= 175.2 MB \end{aligned}$$

Result

- Compressed Size = 168 MB
- Segments = 120,000
- Overhead = 7.2 MB
- Final Data Transmitted = 175.2 MB

Explanation

The Presentation Layer compresses data, the Transport Layer segments it, the Network Layer routes it, the Data Link Layer frames it, and the Physical Layer transmits it.

Application

Healthcare IoT systems require secure, reliable, and efficient communication for patient records and real-time monitoring.

Conclusion

The OSI model enables secure, reliable, and efficient transfer of medical data across healthcare networks.